

Formal Verification of Instability in Randomized Sketch-and-Precondition

Generated from Lean 4 Formalization

Abstract

This document formalizes the instability of the Randomized Sketch-and-Precondition algorithm using a probabilistic argument, based on a Lean 4 formalization. We define the space of 2×2 Subsampled Randomized Hadamard Transform (SRHT) matrices and construct a specific "bad" matrix A . We prove that for any SRHT matrix S drawn from the uniform distribution, the condition number of the preconditioned system is large, demonstrating that the algorithm is not backward stable with probability 1.

1 Definitions and Setup

1.1 Basic Matrices

Definition 1 (Hadamard Matrix). *The Hadamard matrix $H_k \in \mathbb{R}^{2^k \times 2^k}$ is defined recursively:*

$$H_0 = [1]$$

$$H_{k+1} = \begin{pmatrix} H_k & H_k \\ H_k & -H_k \end{pmatrix}$$

Specifically, for $k = 1$:

$$H_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Definition 2 (Sampling and Sign Matrices).

- A matrix $P \in \mathbb{R}^{s \times m}$ is a **sampling matrix** if it selects s rows from the identity matrix (formalized via an embedding $f : \{1 \dots s\} \hookrightarrow \{1 \dots m\}$).
- A matrix $D \in \mathbb{R}^{m \times m}$ is a **sign matrix** if it is diagonal and $|D_{ii}| = 1$ for all i .

Definition 3 (SRHT Matrix). *A matrix $S \in \mathbb{R}^{s \times 2^k}$ is an **SRHT matrix** if there exist a sampling matrix P and a sign matrix D such that:*

$$S = \sqrt{\frac{2^k}{s}} P H_k D$$

1.2 Stability Metrics

Definition 4 (Spectral Norm and Condition Number). *For a matrix A , let $\lambda(A^T A)$ be the set of eigenvalues of $A^T A$.*

- The spectral norm is $\|A\|_2 = \sqrt{\max \lambda(A^T A)}$.
- The condition number is $\kappa(A) = \sqrt{\frac{\max \lambda(A^T A)}{\min \lambda(A^T A)}}$ (if A has full column rank, else 0).

Definition 5 (Instability). *Given parameters $\epsilon, K > 0$, a sketching matrix S , and a target matrix A , the pair (S, A) is **unstable** if there exist a perturbation E , and a QR decomposition $SA + E = QR$ such that:*

1. $\|E\|_2 \leq \epsilon \|SA\|_2$ (The perturbation is small relative to the sketch).
2. $Q^T Q = I$ and R is upper triangular with $\det(R) \neq 0$.
3. $\kappa(AR^{-1}) > K$ (The preconditioned system is ill-conditioned).

2 The 2x2 Construction

We focus on the case where $n = 2^1 = 2$ and the sketch size is $s = 2$.

2.1 The Distribution

Let $\mathcal{U}$ be the uniform probability measure over the set of all valid SRHT matrices for $k = 1, s = 2$.

$$\mathcal{S}_{1,2} = \left\{ \sqrt{\frac{2}{2}} P H_1 D \mid P \in \text{Sampling}(2, 2), D \in \text{Sign}(2) \right\}$$

Note that since $s = n = 2$, the sampling matrices P are effectively 2×2 permutation matrices.

2.2 The "Bad" Matrix A

We construct a specific matrix $A(\epsilon)$ designed to cause instability.

Definition 6 (Bad A).

$$A = \frac{1}{2} H_1 \begin{pmatrix} 1 & 0 \\ 0 & \epsilon \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & \epsilon \\ 1 & -\epsilon \end{pmatrix}$$

3 Structural Lemmas

The proof relies on the specific algebraic structure of SRHT matrices interacting with A .

Lemma 7 (Structure of S). *For any $S \in \mathcal{S}_{1,2}$, S can be decomposed as $S = P H_1 D$ where P is a permutation matrix (either I or $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$) and D is a diagonal sign matrix.*

Lemma 8 (Structure of SA). *For any $S \in \mathcal{S}_{1,2}$, there exist a permutation matrix P' and a sign matrix D' such that:*

$$SA = P' D' \begin{pmatrix} 1 & 0 \\ 0 & \epsilon \end{pmatrix}$$

Proof. (Sketch based on Lean proof `SRHT_mul_bad_A_structure`)

Since $S = P H_1 D$, we analyze $H_1 D A$. Note that $A = \frac{1}{2} H_1 \text{diag}(1, \epsilon)$. Algebraic manipulation of $H_1 D H_1$ shows it is equal to $2 P'' D''$ for some permutation P'' and sign D'' . Thus $SA = P H_1 D (\frac{1}{2} H_1 \text{diag}(1, \epsilon)) = P (\frac{1}{2} H_1 D H_1) \text{diag}(1, \epsilon) = P P'' D'' \text{diag}(1, \epsilon)$. $\square$

Lemma 9 (Gram Matrix of Preconditioned System). *Let $R = \text{diag}(1, \eta)$ where $\eta = \frac{\epsilon}{2(|K|+1)}$. Then:*

$$(AR^{-1})^T (AR^{-1}) = \begin{pmatrix} 1/2 & 0 \\ 0 & 2(|K|+1)^2 \end{pmatrix}$$

Proof. Direct computation. $AR^{-1} = \frac{1}{2} H_1 \text{diag}(1, \epsilon) \text{diag}(1, 1/\eta) = \frac{1}{2} H_1 \text{diag}(1, \epsilon/\eta)$. Using the orthogonality of H_1 (scaled), the Gram matrix diagonalizes to the squares of the singular values. Calculation yields eigenvalues $1/2$ and $2(|K|+1)^2$. $\square$

4 Main Result: Deterministic Instability

Theorem 10 (Instability for all S). *Let $\epsilon, K \in \mathbb{R}$ with $\epsilon > 0$. For any $S \in \mathcal{S}_{1,2}$, the pair (S, A) is unstable according to the definition above.*

Proof. Let $S \in \mathcal{S}_{1,2}$. By the structural lemma, we have $SA = U \begin{pmatrix} 1 & 0 \\ 0 & \epsilon \end{pmatrix}$ where $U = P' D'$ is unitary (orthogonal).

We define the perturbation target $\eta = \frac{\epsilon}{2(|K|+1)}$. We explicitly construct the perturbation E :

$$E = U \begin{pmatrix} 0 & 0 \\ 0 & \eta - \epsilon \end{pmatrix}$$

1. Norm Bound Check: We must show $\|E\|_2 \leq \epsilon \|SA\|_2$. Since U is unitary, $\|E\|_2 = |\eta - \epsilon|$ and $\|SA\|_2 = \max(1, \epsilon)$. We have $|\eta - \epsilon| = \epsilon \left| \frac{1}{2(|K|+1)} - 1 \right|$. For $K \geq 0$, this is less than ϵ . Since $\epsilon > 0$, $\|SA\|_2 \geq 1$ (assuming small ϵ) or generally $\max(1, \epsilon) \geq 1$. Thus $|\eta - \epsilon| \leq \epsilon \leq \epsilon \|SA\|_2$.

2. QR Decomposition: Consider $SA + E$:

$$SA + E = U \begin{pmatrix} 1 & 0 \\ 0 & \epsilon \end{pmatrix} + U \begin{pmatrix} 0 & 0 \\ 0 & \eta - \epsilon \end{pmatrix} = U \begin{pmatrix} 1 & 0 \\ 0 & \eta \end{pmatrix}$$

Let $Q = U$ and $R = \begin{pmatrix} 1 & 0 \\ 0 & \eta \end{pmatrix}$. Since U is orthogonal and R is diagonal (upper triangular), this is a valid QR decomposition.

3. Condition Number Check: We must check $\kappa(AR^{-1}) > K$. Using Lemma 9, the eigenvalues of $(AR^{-1})^T(AR^{-1})$ are $\lambda_{\max} = 2(|K| + 1)^2$ and $\lambda_{\min} = 1/2$.

$$\kappa(AR^{-1}) = \sqrt{\frac{2(|K| + 1)^2}{1/2}} = \sqrt{4(|K| + 1)^2} = 2(|K| + 1)$$

Since $2(|K| + 1) > K$, the condition holds.

Thus, for the chosen E, Q, R , the instability definition is satisfied. $\square$

5 Probabilistic Instability

Theorem 11 (Almost Sure Instability). *For any $\epsilon > 0$ and $K \in \mathbb{R}$, there exists a matrix A such that:*

$$\mathbb{P}_{S \sim \mathcal{U}}[(S, A) \text{ is unstable}] = 1$$

Proof. We choose the matrix A defined in Definition 3. The probability measure is uniform over the finite set $\mathcal{S}_{1,2}$. By Theorem 10, the event $\{S \mid (S, A) \text{ is unstable}\}$ contains the entire support of the distribution. Therefore, the probability of the event is 1. $\square$